SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA5101

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

Total Marks: 50

Annexure A

Code of Conduct:

I **SHAIK ASHA (192473025)** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **SHAIK ASHA**

Annexure A

1. Secure Password Storage in Web Applications
A web application stores user passwords and needs to protect them against database breaches.

A. Explain why storing plaintext passwords is insecure.
B. Suggest a secure method for storing passwords and justify your choice.

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2. Digital Signature for Document Authentication
A company wants to ensure that official documents sent electronically are authentic and unaltered.

A. Explain how digital signatures can be used in this scenario.
B. Justify how this method ensures both integrity and non-repudiation.

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3. Preventing Replay Attacks in Online Transactions
An online payment system is vulnerable to replay attacks where attackers resend previously captured transactions.

A. Explain how replay attacks occur.
B. Suggest a cryptographic technique to prevent such attacks and justify your answer.

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4. Secure Communication over Public Wi-Fi
A user accesses sensitive information using a public Wi-Fi network without encryption.

A. Identify the risks involved in this scenario.
B. Recommend a secure communication method and justify how it protects user data.

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5. Data Integrity in Cloud Storage
An organization stores critical data in a cloud server and wants to ensure that the data is not altered.

A. Explain how hashing can be used to verify data integrity.
B. Suggest an additional mechanism to enhance trust and justify your answer.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Secure Communication over Public Wi-Fi

Introduction

Public Wi-Fi networks are commonly available in places such as airports, railway stations, shopping malls, hotels, libraries, and cafes. They allow users to access the internet without using mobile data. However, public Wi-Fi networks may not provide sufficient security, especially when users access sensitive information without encryption.

Encryption is the process of converting readable information, known as plaintext, into an unreadable form called ciphertext. Only authorized parties with the appropriate decryption key can recover the original information. When a user accesses banking websites, email accounts, or confidential documents through an unsecured connection, attackers may exploit security weaknesses to steal or manipulate information.

Therefore, secure communication methods are necessary to protect confidentiality, integrity, and authenticity while using public Wi-Fi.

A. Identify the Risks Involved in This Scenario

When a user accesses sensitive information over public Wi-Fi without encryption, several security risks arise.

1. Eavesdropping

Eavesdropping occurs when an attacker secretly monitors network communication to obtain sensitive information. On an unsecured network, attackers may be able to capture transmitted data using network-monitoring tools.

For example, if a user sends a password or personal information through an unencrypted connection, an attacker monitoring the network may be able to read it. This can lead to identity theft, financial loss, and unauthorized access to accounts.

2. Man-in-the-Middle Attack

A Man-in-the-Middle (MITM) attack occurs when an attacker positions themselves between two communicating parties, such as a user and a website. The attacker may intercept, read, or modify the information exchanged between them.

For example, a user may believe they are communicating directly with a banking website, while an attacker secretly intercepts the communication. The attacker may attempt to steal login credentials or modify information being transmitted.

Encryption and proper server authentication help prevent such attacks.

3. Data Theft and Identity Theft

Sensitive information transmitted without encryption may be exposed to unauthorized individuals. This information can include usernames, passwords, financial details, personal messages, and confidential business documents.

Attackers may use stolen information to access online accounts, conduct fraudulent transactions, impersonate users, or commit identity theft. The consequences may include financial losses and damage to an individual's reputation.

4. Fake Wi-Fi Hotspots

Attackers may create fraudulent Wi-Fi networks with names similar to those of legitimate networks. These networks are often designed to trick users into connecting to them.

For example, an attacker may create a hotspot named “Free Airport Wi-Fi.” Users who connect to it may unknowingly send their internet traffic through equipment controlled by the attacker.

The attacker may then attempt to monitor traffic, redirect users to fraudulent websites, or collect personal information. Users should verify the network name with the authorized service provider before connecting.

5. Session Hijacking

Session hijacking occurs when an attacker obtains a user's active session identifier, such as a session cookie, and uses it to impersonate the user.

For example, if a session cookie is exposed through an insecure connection, an attacker may attempt to use it to access the victim's account without entering the password again.

HTTPS, secure cookie settings, and proper session management help reduce this risk.

6. Malware and Phishing Attacks

Public Wi-Fi may expose users to malicious websites, fraudulent login pages, and other online threats. Attackers may redirect users to fake websites that resemble legitimate services.

If a user enters their login details on a fraudulent page, the attacker may steal the credentials. Users may also download malicious files that compromise their devices.

Therefore, users should avoid suspicious links, verify website addresses, and keep their devices and applications updated.

B. <u>Recommend a Secure Communication Method and Justify It</u>

Recommended Method: HTTPS/TLS Encryption with a Trusted VPN

The recommended solution is to use websites secured by HTTPS with Transport Layer Security (TLS). For additional protection when connected to public Wi-Fi, users can also use a reputable Virtual Private Network (VPN).

1. HTTPS and TLS Encryption

HTTPS stands for Hypertext Transfer Protocol Secure. It uses TLS to protect communication between a web browser and a website.

The process works as follows:

1. The user opens a website using an HTTPS connection.
2. The browser and server establish a secure TLS connection.
3. The server presents its digital certificate so that the browser can authenticate the server.
4. The browser and server establish the cryptographic keys needed to protect communication.
5. Data exchanged through the connection is encrypted and protected against undetected modification.

Even if an attacker captures the encrypted traffic, they should not be able to read its contents without the necessary cryptographic keys.

HTTPS/TLS therefore provides confidentiality, data integrity, and server authentication. Users should still verify the website address because a fraudulent website can also use HTTPS.

2. Virtual Private Network (VPN)

A VPN creates an encrypted tunnel between a user's device and a VPN server. Internet traffic routed through the VPN tunnel is protected from being read by other users on the local Wi-Fi network.

For example, when a user connects to a trusted VPN before accessing sensitive information at a cafe, the VPN encrypts traffic between the device and the VPN server. This makes local network interception more difficult.

However, a VPN does not automatically make every website safe. Users should continue using HTTPS and select a trustworthy VPN provider.

3. Additional Security Measures

The following measures further improve security:

- **Multi-factor authentication (MFA):** Requires an additional verification factor beyond a password.
- **Strong passwords:** Reduces the risk of unauthorized account access.
- **Software updates:** Fix known vulnerabilities in operating systems and applications.
- **Avoiding suspicious networks:** Reduces the possibility of connecting to fraudulent hotspots.
- **Disabling automatic Wi-Fi connections:** Prevents devices from joining untrusted networks without the user's knowledge.
- **Using mobile data for highly sensitive transactions when appropriate:** Avoids exposure to potentially unsafe public Wi-Fi.

Justification

HTTPS/TLS protects the communication between the user and the intended website. A VPN adds an encrypted layer between the user's device and the VPN server, helping protect traffic on the local network.

Together, these measures significantly reduce the risks of eavesdropping, data interception, and unauthorized modification. Additional precautions, such as MFA and secure browsing practices, provide further protection against account compromise and phishing.

Conclusion

Public Wi-Fi is convenient, but using it without adequate security can expose sensitive information to cyberattacks. Eavesdropping, MITM attacks, fake hotspots, session hijacking, and data theft are major risks. Using HTTPS/TLS, a trusted VPN, MFA, and safe browsing practices provides a more secure communication environment and helps protect users' personal and financial information.

2. Data Integrity in Cloud Storage

Introduction

Cloud storage allows organizations to store, manage, and access data through remote servers over the internet. It is widely used to store financial records, customer information, medical records, business documents, and other critical data.

Although cloud storage offers convenience, scalability, and accessibility, organizations must ensure that stored information remains accurate and unchanged unless an authorized modification is made. Data may become altered because of transmission errors, software failures, unauthorized access, or malicious attacks.

Data integrity refers to the property of data remaining accurate, complete, and consistent throughout its storage, transmission, and use. Cryptographic hashing is an important technique for checking data integrity. Additional mechanisms, such as digital signatures and access controls, can strengthen the reliability and trustworthiness of cloud storage.

A. Explain How Hashing Can Be Used to Verify Data Integrity

1. Meaning of Hashing

Hashing is a cryptographic process that converts input data of any size into a fixed-length output called a hash value, hash code, or message digest.

A cryptographic hash function takes a file or message as input and produces a digest that represents the contents of that data. Common hashing algorithms include SHA-256 and SHA-3. MD5 and SHA-1 are older algorithms and should not be selected for new security-sensitive integrity systems because of their known weaknesses.

A secure cryptographic hash function should have the following properties:

- **Deterministic:** The same input always produces the same hash.
- **Fixed-length output:** The hash has a defined length regardless of the input size.
- **One-way property:** It should be computationally infeasible to recover the original data from its hash.
- **Collision resistance:** It should be computationally infeasible to find two different inputs that produce the same hash.
- **Sensitivity to changes:** Even a small change in the input should produce a substantially different hash with overwhelming probability.

For example, SHA-256 produces a 256-bit hash value, usually represented as 64 hexadecimal characters.

2. Procedure for Verifying Data Integrity

An organization can use hashing to check whether a cloud-stored file has changed.

Step 1: Generate the Original Hash

Before uploading a critical file, the organization calculates its hash using a secure algorithm such as SHA-256.

For example, suppose a company has a confidential financial report. The organization calculates the SHA-256 hash of the original report.

Step 2: Store the Original Hash Securely

The original hash is stored in a trusted location, separate from the file being checked. Access to the stored hash should be restricted to authorized users.

This is important because an attacker who can replace both the file and its reference hash may be able to conceal unauthorized changes.

Step 3: Upload the File to the Cloud

The organization uploads the report to the cloud server. The file may then be accessed, downloaded, or transferred when required.

Step 4: Calculate the Hash Again

When the organization wants to verify the report, it retrieves the file and calculates a new SHA-256 hash using the same algorithm.

Step 5: Compare the Hash Values

The newly calculated hash is compared with the original hash.

- If both hash values match, the file is consistent with the original data with very high confidence.
- If the hash values differ, the file may have been modified, corrupted, or damaged during storage or transmission.

3. Advantages and Limitations of Hashing

Advantages:

1. Hashing provides an efficient way to detect changes in stored files.
2. It can verify data after uploading, downloading, or transferring files.
3. It requires relatively little storage space because the hash has a fixed length.
4. It can be used with large files and automated integrity-checking systems.
5. It helps detect accidental corruption and unauthorized modifications.

Limitations:

Hashing alone does not prevent a file from being modified. It only helps detect changes when the reference hash is trustworthy. A basic hash also does not prove who created the file or who made a modification.

Therefore, organizations should protect reference hashes and use additional security mechanisms, such as digital signatures, message authentication codes, access controls, and audit logs.

B. Suggest an Additional Mechanism to Enhance Trust and Justify Your Answer

Recommended Mechanism: Digital Signatures

Digital signatures are a cryptographic security mechanism that helps verify the authenticity and integrity of digital information. They use public-key cryptography and are particularly useful when an organization needs to confirm that a file came from an authorized source and has not been modified after signing.

A digital signature is created using the signer's private key and verified using the corresponding public key.

1. Working of Digital Signatures

Step 1: Generate a Hash

The organization calculates the cryptographic hash of the original file using SHA-256.

Step 2: Create the Digital Signature

The organization uses its private signing key and a suitable digital-signature algorithm to sign the data, typically through a cryptographic signature process involving the file's hash.

Step 3: Store the File and Signature

The organization stores the file and its digital signature in the cloud. The corresponding public key or a trusted digital certificate is made available to the verifier.

Step 4: Retrieve the File

When an authorized user downloads the file, the user obtains the associated digital signature.

Step 5: Verify the Signature

The recipient verifies the digital signature using the signer's public key and the appropriate verification algorithm.

If verification succeeds, it provides evidence that the signed data has not been modified since signing and that the signature was produced using the corresponding private key.

The public key must be reliably associated with the organization; otherwise, an attacker could substitute a different public key and undermine authentication.

2. Advantages of Digital Signatures

Data integrity: Digital signatures detect unauthorized changes to signed data.

Authentication: They help verify that a signature was produced using the private key associated with the claimed signer.

Non-repudiation: They provide evidence that a particular key was used to sign the data. This assurance depends on secure private-key management, reliable identity verification, and appropriate procedures.

Greater trust: Digital signatures help protect against attempts to replace a cloud file with modified data while presenting it as an authentic document.

Support for auditing: Verified signatures can help organizations establish the authenticity of important documents during audits and investigations.

3. Additional Measures to Strengthen Cloud Data Integrity

Digital signatures can be combined with the following controls:

- **Access control:** Restrict file modification permissions to authorized users.
- **Multi-factor authentication:** Strengthen authentication for accounts with access to critical information.
- **Audit logs:** Record who accessed, modified, uploaded, or deleted files and when those actions occurred.
- **Versioning and backups:** Maintain previous file versions and recover data following accidental or malicious changes.
- **Key management:** Protect private signing keys and limit access to authorized personnel.
- **Integrity monitoring:** Recalculate and compare file hashes periodically to identify unexpected changes.

These measures provide multiple layers of protection rather than relying on a single security technique.

Justification

Hashing is useful for detecting changes, but it cannot independently prove who created a file or whether the reference hash is trustworthy. Digital signatures add authenticity verification and provide stronger evidence that the signed data has not been altered.

For example, if an attacker modifies a company's financial report in cloud storage, the signature verification process should fail because the modified data no longer matches the signed content. The organization can then reject the file and investigate the incident.

Combining SHA-256 hashing, digital signatures, access controls, audit logs, and backups provides a stronger solution for maintaining data integrity and trust in cloud storage.

Conclusion

Data integrity is essential for organizations that store critical information in cloud environments. Cryptographic hashing allows an organization to detect changes by comparing the hash of a retrieved file with a securely stored reference hash. However, hashing alone does not establish the identity of the data's creator or prevent unauthorized modifications.

Digital signatures strengthen the solution by verifying the authenticity and integrity of signed information. When combined with access controls, audit logs, secure key management, and backups, these mechanisms help organizations protect important data and maintain confidence in cloud storage systems.